



ESOFT Metro Campus

School of Computing

Bachelor of Information Technology (Honours)

Batch 07 – Level 01 – Semester 01

CCU2304 Information Systems in Organizations

Assignment 01

Name: M.W.Oshadhi Chathurani Wickramasinghe.

Registration Number: E310875

Module Lecture: Ms. Lasanthika.

Submission Date: 28/07/2026

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Introduction

In today's fast-paced business environment, organizations rely heavily on how efficiently they manage their daily operations and share information across different departments. Information Systems (IS) act as the vital backbone for modern businesses, helping them integrate data smoothly (Laudon and Laudon, 2022), reduce communication delays, and gain a strong competitive edge. This report explores how Mount Lavinia Hotel, a premier heritage beachfront hotel in Sri Lanka, utilizes information systems to support its functional areas, assist management in decision-making, and secure critical corporate assets (Mount Lavinia Hotel, 2023). Furthermore, the report examines the role of digital infrastructure in optimizing guest services, streamlining housekeeping and front office functions, and addressing modern cyber security challenges within the hospitality sector. In the modern hospitality industry, the integration of advanced Information Systems is no longer merely an operational luxury, but a fundamental prerequisite for sustained competitive advantage and operational excellence. Hotels face continuous pressure to balance traditional heritage with cutting-edge technological advancements to meet evolving customer expectations. For historic establishments like Mount Lavinia Hotel, maintaining personal charm while streamlining back-end logistics, customer relationship management, and financial reporting requires a robust digital architecture. Information systems bridge this gap by facilitating seamless data flow across departments, reducing manual errors, and empowering managers at all levels to make swift, data-driven decisions in a dynamic market environment.



Figure 2: Mount Lavinia Hotel Overview



Figure 1: Mount Lavinia Hotel Overview

Activity 01: Organization Overview and Information Systems Usage

a. Overview of the Selected Organization.

For this assignment, we examine Mount Lavinia Hotel, a historic beachfront resort situated just south of Colombo (Mount Lavinia Hotel, 2023).

- **Brief History and Background:**

- Originally built in the 1800s as the residence of British Governor Sir Thomas Maitland, the property has evolved into a renowned 275-room heritage hotel blending colonial charm with modern hospitality standards. Over the decades, the establishment has successfully preserved its historical architectural identity while expanding its capacity to accommodate international tourists, corporate delegates, and grand celebratory events. This unique blend of heritage preservation and contemporary service excellence forms the foundational bedrock of the organization's long-term business strategy.

- **Vision:**

- To remain a leading heritage beachfront resort celebrated for timeless elegance and exceptional service. By continuously aligning luxury hospitality with innovative digital management solutions, the resort aims to set a benchmark for sustainable tourism and guest satisfaction within the South Asian region.

- **Mission:**

- To deliver outstanding lodging, dining, and banqueting experiences by preserving rich history while adopting modern service standards. This is achieved by empowering staff through advanced information systems, fostering eco-friendly operational practices, and ensuring every guest enjoys a seamless blend of historical ambiance and contemporary convenience.

- **Core Business Objectives:**

- Maintaining high year-round room and banquet occupancy rates. Through dynamic pricing strategies and targeted digital marketing campaigns.
- Preserving cultural heritage architecture while integrating modern digital guest experiences. Ensuring structural authenticity coexists harmoniously with high-speed wireless networks and automated mobile check-ins.

- Maintaining excellent customer service standards across all dining, leisure, and event sectors. Backed by real-time customer feedback analysis and comprehensive staff training modules.
- **Organizational Structure and Key Departments:**

The hotel is managed through a formal hierarchy led by the General Manager, supported by specialized functional areas designed to ensure smooth business operations.

- **Front Office:** Manages guest relations, room reservations, check-ins, and billing inquiries. It acts as the primary nerve center for guest communications, ensuring that individual preferences and special requests are accurately recorded and communicated across departments to deliver a personalized hospitality experience.



Figure 4: Guest Room View



Figure 3: Guest Room View

- **Housekeeping and Maintenance:** Ensures strict room cleanliness, linen care, and the upkeep of historic facilities. Given the heritage nature of the property, this department balances modern sanitation protocols with specialized care for colonial-era architecture and interior structures.
- **Food and Beverage/Kitchen:** Oversees restaurants, bars, beachside dining, and large wedding banquets (Mount Lavinia Hotel, 2023). By coordinating closely with suppliers and event planners, the team ensures high culinary standards and seamless execution for both daily dining and large-scale celebrations.
- **Human Resources:** Manages workforce-related activities including recruitment, training, maintaining employee records, and handling compliance with labor laws. It also conducts continuous digital skill development programs to ensure staff members are proficient in utilizing newly integrated hotel information systems.
- **Account and Finance:** Manages company funds, produces invoices, tracks payments, manages budgets, and prepares financial reports such as cash flow forecasts. This

department relies heavily on integrated financial management systems to maintain real-time visibility over revenues generated from rooms, dining, and banquets.

b. Functional Areas and Information Systems Usage.

To ensure smooth daily operations, Mount Lavinia Hotel divides its functions into specialized departments, each supported by information systems:

- **Accounts and Finance:**
 - Manages capital funds and financial planning by producing invoices, tracking payments, managing budgets, and preparing financial statements using automated financial software integrated with Point-of-Sale (POS) systems.
- **Human Resource (HR) Management:**
 - Handles workforce planning, employee recruitment, training schedules, and labor law compliance by employing Human Resource Information System and payroll systems to track staff attendance and automate salary processing.
- **Sales and Marketing:**
 - Drives revenue through product and service sales by promoting offerings, responding to customer inquiries, managing sales, generating price quotations, and updating customer databases and billing records (Mount Lavinia Hotel, 2023).
- **Operations (Front Office and Inventory):**
 - Ensures smooth business processes and communication by handling meeting schedules, office documents, purchasing supplies, and coordinating company events.

Activity 02: Management Levels and Major Information Systems Analysis.

a. Support for Decision-Making at Management Levels.

At Mount Lavinia Hotel, information systems provide timely, accurate, and relevant information to assist managers across different organizational tiers in planning, controlling, and problem-solving:

Decisions taken in an Org:

- There are three types of decisions taken in an organization

1. Strategic Decision,
2. Tactical Decision,
3. Operative Decision.



Figure 5: Management Levels and Decision Types

- **Operational Level (Bottom Management):**

- Operating managers and self-directed teams rely on detailed, frequent, and internal task-specific data (Laudon and Laudon, 2022). Systems like Transaction Processing Systems (TPS) provide real-time updates on daily room check-ins, guest billing logs, and transaction-oriented applications (OLTP) to ensure smooth day-to-day execution.

- **Tactical Level (Middle Management):**

- Middle-level managers handle semi-structured decisions regarding department plans, policies, and supplier choices. They utilize Management Information Systems (MIS) and Decision Support Systems (DSS) to review structured periodic summary reports, compare seasonal sales, and analyze complex problems using mathematical models and simulations (Laudon and Laudon, 2022).

- **Strategic Level (Top Management):**

- Executives and directors make complex, long-term, unstructured decisions. Using Executive Information Systems (EIS), they access a high-level bird's-eye view of the organization through user-friendly dashboards and external market trends to evaluate long-term vision, goal setting, and major business strategies (Laudon and Laudon, 2022).

b. Analysis of a Major Information System

- **Purpose of the Selected System:**

- The hotel utilizes an integrated Property Management System (PMS) (acting as a core transaction and management tool) to centralize room reservations, guest history, front office operations, and billing into a single unified platform. Furthermore, it acts as the digital backbone that eliminates data silos between departments, ensuring seamless coordination from guest arrival to departure.

- **Main Components and Features:**

- Features include online booking engines, real-time room status tracking, guest profile databases, and Point-of-Sale (POS) billing modules. These modules allow staff to instantly update room availability, track dining charges, and maintain accurate records of recurring guest preferences.

- **Types of Information Processed:**

- It handles internal transactional data (daily room occupancy, laundry requests, and kitchen inventory levels) and external data (tourist arrival trends, online booking inquiries). This dual data processing capability ensures that management can monitor both immediate floor-level activities and broader market shifts.

- **How it Improves Operational Efficiency:**

- It eliminates manual paperwork, prevents double-booking errors, automates routine tasks, and speeds up guest check-in times through real-time data processing. Consequently, staff members spend less time on repetitive administrative burdens and can focus more on delivering exceptional guest service.

- **How it Supports Managerial Decision-Making:**

- It generates scheduled periodic reports (like daily revenue summaries) and ad-hoc reports that help managers analyze peak booking seasons and manage staff rostering effectively.

- **Benefits and Limitations:**

- **Benefits:** High operational accuracy, reduced manual errors, secure data storage, and enhanced customer satisfaction through real-time services. Furthermore, integrated systems facilitate seamless cross-departmental communication, allowing management to track overall performance metrics and respond promptly to dynamic market demands (Laudon and Laudon, 2022).
- **Limitations:** High software installation costs, staff training requirements, and vulnerability to system downtime if technical failures occur. Additionally, legacy system integration issues and potential resistance to change from veteran staff members can occasionally slow down the digital transition process.

Activity 03: Operational Issues and Cloud-Based Information System Solution

a. Identify Operational Issue and Propose an Information System Solution

- **Operational Issue:**
 - Mount Lavinia Hotel often experiences communication delays between the Front Office and the Housekeeping department. When a guest checks out, manual communication or delayed phone calls cause housekeepers to clean rooms late, resulting in long waiting queues for newly arriving guests.
- **Proposed Information System Solution:**
 - We propose upgrading to a Real-Time Cloud-Based Property Management System (PMS) integrated with mobile tablets for the housekeeping team. The moment a guest checks out at the front desk, the system instantly updates the room status on the housekeepers' mobile devices, ensuring fast room turnaround. Furthermore, implementing this cloud-based mobile solution allows management to monitor housekeeping Key Performance Indicators (KPIs) in real-time, generate automated staff efficiency reports, and drastically reduce guest check-in waiting times, thereby maximizing overall operational productivity (Laudon and Laudon, 2022).



Figure 6: Real-Time Property Management System Interface

b. Evaluation against Information Requirements & Decision-Making

The proposed cloud-based system effectively addresses information needs across all management levels:

- **Operational Level:**
 - Gives front-line staff and room attendants instant, real-time lists of tasks, completely removing manual tracking delays. Additionally, it ensures that daily transaction logs and guest check-in details are updated without errors. This enables front-office personnel to handle daily guest interactions smoothly and maintain high accuracy in record-keeping (Laudon and Laudon, 2022).

- **Tactical Level:**
 - Generates periodic summary reports (such as average room cleaning times and staff productivity metrics) for middle managers to optimize shift scheduling. This helps department heads monitor resource allocation and control operational costs efficiently and supports data-driven decision-making across department (Laudon and Laudon, 2022).
- **Strategic Level:**
 - Provides senior management with aggregated performance insights and historical data to forecast future capacity needs during peak tourism seasons. Consequently, executives can make well-informed, long-term decisions regarding business expansion and technology investments while maintaining a competitive advantage in the industry.

Activity 04: Information Security Challenges and Mitigation Strategies.

a. Information Security Issue and Solution

- **Security Issue:**
 - Mount Lavinia Hotel handles a large volume of sensitive international guest data and credit card payments. A major security challenge is the risk of unauthorized access or a ransom ware attack where malicious software encrypts the organization's file servers and demands a ransom for the decryption key, halting business operations and potentially damaging corporate reputation.
- **Proposed Solution:** Implement the CIA Triad framework alongside strong administrative, physical, and technical controls to protect critical corporate assets (Laudon and Laudon, 2022):
 - Confidentiality: Enforce robust file encryption, strict user access permissions, and Multi-Factor Authentication (MFA) so only authorized personnel can access sensitive credit card and project files.
 - Integrity: Use file integrity monitoring and hash validation to ensure guest booking, document, and payment records cannot be secretly altered or tampered with by unauthorized modifications.
 - Availability: Set up automated daily cloud backups and a robust disaster recovery plan to guarantee that booking systems remain online and files can be quickly restored without paying a ransom during technical failures or cyber threats.

b. Justification of Proposed Solution

The proposed security enhancements bring significant value to the organization:

- **Information Security:**

- Effectively shields confidential guest records and financial transactions from cyber-attacks, preventing severe data breaches, direct economic loss, and legal consequences. Furthermore, it ensures adherence to industry standard compliance requirements regarding digital data protection.
- **Operational Reliability:**
 - Automated backups and disaster recovery protocols ensure that reservation and file services do not experience extended downtime during system failures, minimizing business disruption. Consequently, staff members can maintain service delivery without unexpected technical interruptions.
- **Stakeholder Trust:**
 - Demonstrating strict commitment to data privacy reassures international guests and corporate partners, preventing the loss of investor trust and building long-term brand loyalty. This proactive approach strengthens the hotel's market reputation as a secure and reliable hospitality provider.

- **Operational Reliability:**

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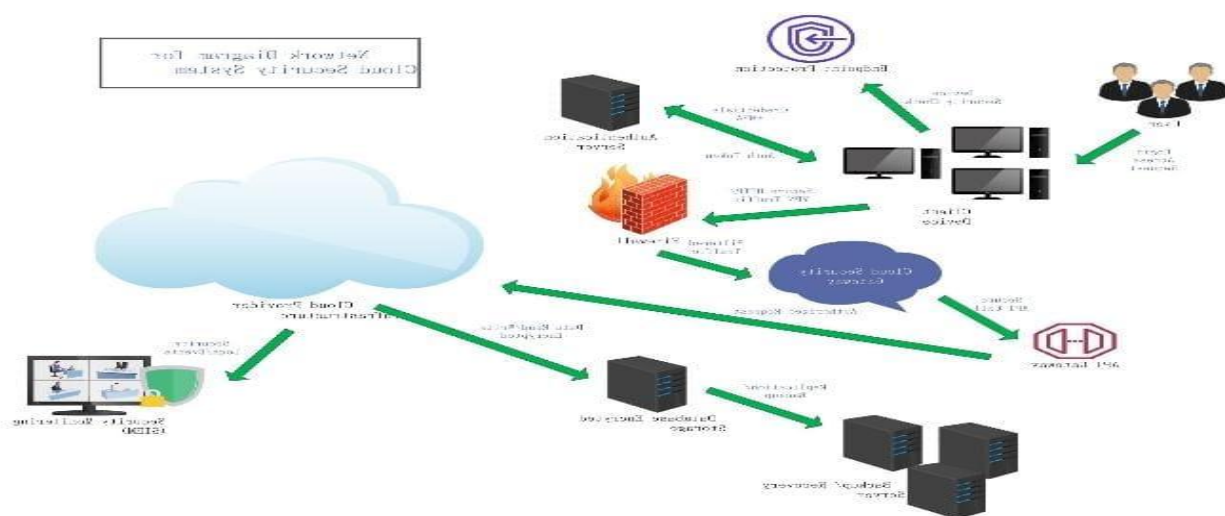


Figure 7: Cloud Security and Network Architecture

Conclusion

a. Summary of Information Systems Integration

In conclusion, the strategic implementation and integration of information systems across Mount Lavinia Hotel have demonstrated a profound impact on overall operational efficiency and management decision-making. From utilizing Transaction Processing Systems (TPS) at the operational level for daily room check-ins and billing, to leveraging Management Information Systems (MIS) and Decision Support Systems (DSS) at the tactical level for shift scheduling and cost control, technology acts as the backbone of the organization (Laudon and Laudon, 2022).

Furthermore, Executive Information Systems (EIS) empower senior leadership to analyze long-term market trends and maintain a competitive edge in Sri Lanka's thriving hospitality sector.

b. Final Organizational Outlook

Despite encountering minor operational hurdles such as cross-departmental communication delays and cyber security vulnerabilities, the proposed cloud-based solutions and robust administrative controls-such as the CIA Triad framework-provide a secure pathway for sustainable growth. By continuously upgrading digital tools, training front-line and administrative staff, and maintaining a customer-centric digital approach, Mount Lavinia Hotel successfully preserves its historic heritage while embracing modern technological advancements (Mount Lavinia Hotel, 2023). Furthermore, the successful integration of digital platforms within the hospitality sector heavily relies on continuous staff adaptability and a proactive management approach. As guest expectations rapidly shift toward contactless services, instant booking updates, and personalized digital experiences, hotels must consistently evaluate their technological infrastructure. Investing in secure, scalable, and user-friendly information systems not only streamlines day-to-day administrative burdens but also ensures long-term business resilience in a competitive global market.

Recommendations

- **Enhancing Cybersecurity Infrastructure:**

- Mount Lavinia Hotel should consistently update its firewall protections, conduct routine system vulnerability assessments, and implement mandatory cybersecurity awareness training for all administrative and front-office personnel to prevent potential ransomware threats.

- **Investing in Cloud Scalability:**

- To handle fluctuations in international guest traffic smoothly, the management should upgrade its cloud infrastructure, ensuring seamless data synchronization across reservation, billing, and housekeeping departments without system lags.

- **Upgrading Staff Digital Literacy:**

- Regular workshops and technical training sessions should be organized for employees' at all operational levels to maximize the efficiency of newly integrated Information Systems and minimize human entry errors.

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